

Gap C1 Closure: P_ψ -Odd Charged Current

Chirality from $P_\psi = \gamma^5$ [[L1]] and $SU(2)_L$ left action [[L1 cond.]]

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2026-06-14

Abstract

Gap C1 required showing the $W^\pm$ vertex is P_ψ -odd from $S[\Theta]$. We close it by showing the *charged current* J_W^μ is P_ψ -odd, which follows from: (i) $P_\psi = \gamma^5$ [[L1], `alpha_conditions_closure.tex`], so left-handed fermions are P_ψ -odd and right-handed are P_ψ -even; (ii) $SU(2)_L$ = left multiplication on Θ [[L1 cond.], T2_GAUGE], which couples exclusively to P_ψ -odd (left-handed) fermions. The $W^\pm$ gauge field itself is P_ψ -even (from P_ψ invariance of $S[\Theta]$), but the charged current $J_W^\mu = W_\mu^+ \bar{\psi}_R \gamma^\mu \psi_L$ is P_ψ -odd: $P_\psi(J_W^\mu) = -J_W^\mu$. Gap C1 is **CLOSED** [[L1 cond.] on T2_GAUGE].

1 P_ψ Parity of the $W^\pm$ Gauge Field

Proposition 1.1 (W_μ^a is P_ψ -even — [L1]). *The $SU(2)_L$ gauge field W_μ^a is P_ψ -even: $P_\psi(W_\mu^a) = +W_\mu^a$.*

Proof. The covariant derivative $D_\mu \Theta = \partial_\mu \Theta - iW_\mu^a [\sigma^a/2, \Theta]$ must transform consistently under P_ψ . Since $P_\psi \Theta_n = (-1)^n \Theta_n$ [[L0]] and $S[\Theta]$ is P_ψ -invariant [[L1], `step1_psi_parity.tex`]:

$$P_\psi(D_\mu \Theta_n) = \partial_\mu((-1)^n \Theta_n) - iW_\mu^a [\sigma^a/2, (-1)^n \Theta_n] = (-1)^n D_\mu \Theta_n.$$

This is consistent with $P_\psi(W_\mu^a) = +W_\mu^a$. [L1] □

2 P_ψ Parity of the Charged Current

Theorem 2.1 (Charged current is P_ψ -odd — Gap C1 CLOSED — [L1 cond.]). *The $W^\pm$ charged current*

$$J_{\text{charged}}^{\mu, W} = W_\mu^+ \bar{\psi}_R \gamma^\mu \psi_L + W_\mu^- \bar{\psi}_L \gamma^\mu \psi_R$$

is P_ψ -odd: $P_\psi(J_{\text{charged}}^{\mu, W}) = -J_{\text{charged}}^{\mu, W}$.

Proof. Step 1 [[L1]]. From $P_\psi = \gamma^5$ [`alpha_conditions_closure.tex`, Theorem 1]:

$$\begin{aligned} P_\psi \psi_L &= -\psi_L & (\text{left-handed: } \gamma^5\text{-eigenvalue } -1), \\ P_\psi \psi_R &= +\psi_R & (\text{right-handed: } \gamma^5\text{-eigenvalue } +1). \end{aligned}$$

Step 2 [[L0]]. Bilinear $\bar{\psi}_R \gamma^\mu \psi_L$ transforms as:

$$P_\psi(\bar{\psi}_R \gamma^\mu \psi_L) = \overline{(P_\psi \psi_R)} \gamma^\mu (P_\psi \psi_L) = (+\bar{\psi}_R) \gamma^\mu (-\psi_L) = -\bar{\psi}_R \gamma^\mu \psi_L.$$

Step 3 (Proposition ??). $P_\psi(W_\mu^+) = +W_\mu^+$.

Combining:

$$P_\psi(W_\mu^+ \bar{\psi}_R \gamma^\mu \psi_L) = (+W_\mu^+) (-\bar{\psi}_R \gamma^\mu \psi_L) = -W_\mu^+ \bar{\psi}_R \gamma^\mu \psi_L.$$

Similarly for the W^- term. Therefore:

$$P_\psi(J_{\text{charged}}^{\mu,W}) = -J_{\text{charged}}^{\mu,W}. \quad [\mathbf{L1 \ cond.}]$$

Source of [L1 cond.]: $SU(2)_L$ couples to ψ_L and not ψ_R by the T2_GAUGE result that $SU(2)_L$ acts from the left on the spinorial component of Θ [[L1 cond.], T2_GAUGE paper]. $\square$

Corollary 2.2 (OP-S4 closed — [L1 cond.]). *The P_ψ -odd charged current selects left-handed fermions only. $SU(2)_R$ coupling to ψ_L would require a P_ψ -even charged current, which is absent. All three OP-S4 loopholes are closed:*

1. P_ψ -even $SU(2)_R$ coupling to $n < 0$ sector: absent (matter in $n > 0$ by Lemma 4 of Step 1).
2. $SU(2)_R$ coupling via P_ψ -even bilinear: absent (Theorem ??).
3. Accidental P_ψ breaking: excluded by P_ψ -invariance of $S[\Theta]$ [[L1]].

Status of OP-S4: [L1 cond.] on T2_GAUGE. Gap C1: **CLOSED**.

3 T2_GAUGE Status After This Note

T2_GAUGE open conditions — updated.

Condition	Level	Status
$SU(3) \times SU(2)_L \times U(1)_Y$ from $\mathbb{C} \otimes \mathbb{H}$	[L1]	Closed
Three generations from ψ -winding	[L0]	Closed
$\sin^2 \theta_W = 3/8$ at GUT scale	[L1] cond.	Closed
$Y = (B - L)/2$ hypercharge	[L1 cond.] on OP-S4+SU(3)	Closed (C2 Step 1)
OP-S4 chirality: $SU(2)_L$ not $SU(2)_R$	[L1 cond.]	Closed here
Gap C1: P_ψ -odd $W^\pm$ vertex	[L1 cond.]	Closed here
$SU(2)_L = \text{left action on } \Theta$	[L1 cond.]	Sole remaining (T2_GAUGE core result)
C2-ii: $U(1)_B$ from ψ -winding	[Open/MC]	Does not block paper
O3: Higgs/Yukawa	[Open]	Does not block paper

T2_GAUGE is [[L1] cond.] on a single core result: $SU(2)_L$ is the symmetry of left multiplication on $\Theta \in \mathbb{C} \otimes \mathbb{H}$. This is the *defining* result of T2_GAUGE — it is not a gap, it is the main theorem.

Logical structure: T2_GAUGE and the alpha chain share the single remaining condition ($SU(2)_L$ left action). Both are promoted to unconditional **[[L1]]** simultaneously once $SU(2)_L = \text{left multiplication}$ is established as a canonical theorem without bridge assumptions.

What “ $SU(2)_L = \text{left action}$ ” means.

In $\mathbb{C} \otimes \mathbb{H} \cong \text{Mat}(2, \mathbb{C})$, left multiplication $\Theta \mapsto U_L \Theta$ for $U_L \in SU(2)$ is the natural left-action. The T2_GAUGE paper shows this generates the observed $SU(2)_L$ weak interaction. This is the core derivation of the paper.

It is **[[L1 cond.]]** in the sense that it uses the identification of Θ with the SM fields (the UBT-to-SM dictionary). Once this dictionary is established as canonical (= unconditional **[[L1]]**), the entire T2_GAUGE paper and the alpha chain become unconditional **[[L1]]**.

The dictionary is the standard UBT bridge input shared by T1_GR, T2_GAUGE, and T3_ALPHA.

References

- [1] D. Jaros, `canonical/chirality/step1_psi_parity.tex`, 2026.
- [2] D. Jaros, `canonical/chirality/step4_no_wr_derivation.tex`, 2026.
- [3] D. Jaros, `alpha_conditions_closure.tex`, 2026.
- [4] D. Jaros, `papers/UBT_Gauge_Submission.tex`, 2026.
- [5] D. Jaros, `research_tracks/EW/hypercharge_from_ubt.tex`, 2026.